

DISTRIBUTED AND CLOUD COMPUTING

LAB 11: SPARK FUNDAMENTALS & ADVANCED TRANSFORMATIONS

Task 1: Spark SQL

TASK DESCRIPTION: Implement the examples from last week using spark.SQL

DATASET: Pokemon.csv

TASK 1: Compute the total number of Pokemons for each pokemon type (only consider primary type 'Type 1')

TASK 2: Compute the total number of Legendary pokemon for each type

TASK 3: Compute the names of the fastest pokemon for:

- Only considering the primary type
- Considering both primary and secondary type

Spark SQL uses a very standard definition of SQL. You can checkout the available SQL STATEMENTS in the following link:

<https://spark.apache.org/docs/latest/sql-ref-syntax.html>

Task 2: Window Functions

TASK DESCRIPTION: In the commerce dataset utilise the window function to calculate the total number of items sold per hour

DATASET: Commerce.csv

Hints:

- Utilise the window example provided to help you
- Check the data types to ensure they are compatible with window

Task 3: Combine UDFs, Explode & Windows

TASK DESCRIPTION: Utilise UDFs, Exploding and Windows to calculate the number of people at work each hour

DATASET: work_arrival.csv

Hints:

- Utilise the examples for UDF, exploding and window to guide you
- Check the data types to ensure they are compatible with window